

HEMANT SINGH RANA

FULL-STACK SOFTWARE ENGINEER

Dehradun 📍 +91-9082712612 ✉ iamhemant1289@gmail.com [in linkedin.com/in/hemant-singh](https://www.linkedin.com/in/hemant-singh)
github.com/hemant1883

Summary

Dynamic **Full-Stack Software Engineer** specializing in backend development using **Java 21**, **Spring Boot 3**, and **React.js**. Experienced in designing scalable **RESTful APIs**, implementing secure **JWT-based authentication**, developing database-driven applications using **Spring Data JPA** and **Hibernate**, and building responsive web interfaces. Passionate about applying **Clean Architecture** and object-oriented design principles to develop scalable, maintainable software solutions.

Technical Skills

Languages: Java (17/21), C++, JavaScript (ES6+), SQL (MySQL), HTML5/CSS3

Backend Frameworks: Spring Boot 3, Spring Security, Hibernate ORM, RESTful APIs, Spring Data JPA

Frontend Frameworks: React.js, Tailwind CSS

Tools & Concepts: Git, GitHub, Docker, JWT Authentication, Redis, Postman, System Design, Object-Oriented Programming (OOP)

Projects

🔗 College Management System (Full-Stack Web Application) | *Java, Spring Boot, React.js, MySQL* **2026**

- Architected a high-concurrency management platform supporting Students, Faculty, and Admin using **Java 21 Virtual Threads** for improved scalability and performance.
- Engineered an **Automated Faculty Status Engine** that reduced scheduling conflicts by 50% through real-time timetable scanning and stateless session management using **Spring Security**.
- Developed a dynamic exam seating visualization module using **React.js** and **Tailwind CSS**, transforming relational database records into an interactive graphical UI.
- Improved application responsiveness by 40% through efficient client-side polling and centralized global exception handling for reliable real-time data synchronization.
- Applied Clean Code principles and DTO patterns to decouple the persistence layer from the RESTful API layer, reducing payload size by 20%.

🔗 StayEase — Hotel Booking Web Application | *Java, Spring Boot, React.js, MySQL, JWT, Railway* **2026**

- Developed a scalable hotel reservation platform with secure **Role-Based Access Control (RBAC)**, **JWT-based authentication**, and RESTful APIs for user and booking management.
- Designed and implemented JPA entity relationships to manage hotel inventory efficiently while preventing race conditions during overlapping booking requests.
- Optimized MySQL database performance by 30% through efficient indexing and custom SQL query optimization for complex availability searches.
- Streamlined deployment using automated deployment workflows on **Railway**, ensuring 99.9% uptime for the production environment. [\[Live Link\]](#)

Education

Bachelor of Technology (B.Tech.) in Computer Science and Engineering **Aug 2023 – Ongoing**
Swami Rama Himalayan University, Dehradun | CGPA: 7.5 | NAAC A+ Grade

Intermediate (XII) **May 2023**
Holy Angel Sr. Sec. School, Uttarakhand | Percentage: 82%

Achievements & Certifications

- **Winner** – Smart India Hackathon (SIH) 2025 Internal Ideathon (Ranked 1st among 50+ teams)
- **Top 50 Finalist** – SAARTHI National Level Hackathon
- **Certification** – Full-Stack Java Development (Spring Boot & React)